Avi Gobrin

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Education

University of Toronto 2023 – Apr 2027
Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor *Toronto, ON*

Relevant Coursework: Probability & Statistics I/II, Methods of Data Analysis, Multivariate Data Analysis, Statistical Computation, Nonlinear Optimization, Mathematical Modelling, Algorithm Design & Analysis, Data Structures, Software Design, Theory of Computation, Real Analysis, Advanced Calculus, Graph Theory, Combinatorics

Research

University of Toronto, Department of Statistical Sciences Position begins Fall 2026
Incoming Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute) *Toronto, ON*

- Preparing research on the connection between in-context learning and fine-tuning, characterizing both as low-rank weight updates.

Technical Reports

- Engagement, Performance, and Student Outcomes at the Open University (n = 23,301) [Report]
- Discovering Latent Structure in the IPIP Big Five Personality Inventory (n = 19,718) [Report]
- What Makes a Wine Good? Chemical Structure and Quality in Portuguese Wines (n = 6,497) [Report]

Experience

Nextdoor May 2026 – Present
Machine Learning Intern *San Francisco, CA*

- Increased growth and weekly active users by launching Best Places for Families, an SEO ranking product covering 300,000 US neighbourhoods, building every layer end to end: internal data and Census/ACS ingestion through Airflow pipelines to a Postgres/GraphQL/React serving path.
- Compared 4 machine learning models OLS, ridge, NNLS, and monotone GBM across 197,556 neighbourhoods to design the scoring methodology, predicting family-friendlyness scores. Screened 33 candidate metrics down to 8 features. Validated rankings against a preregistered neighbourhood adversarial test set with pairwise Bradley-Terry judgments, cross-checked by both LLM and manual review.
- Made the system self-sustaining with scheduled jobs, tests, and monitoring, and built Databricks tooling that lets non-technical stakeholders retune feature weights and see how any city ranks.

Shopify May – Aug 2025
Software Engineering Intern *Toronto, ON*

- Increased merchant activation and retention by building a full-stack partner-attach system (Ruby on Rails, Calendly API) that screens each new merchant and books qualifying ones into guided onboarding with a vetted advisor. Routing over 100 merchants per day into the partner attach flow.
- Brought Shopify's entire German merchant base into VAT compliance in the 1.5 months from regulatory notice to launch. Reconciled what data Shopify already held against what merchants had to supply, integrated a difficult government tax API, and built a Rails intake form with an email flow that returns each merchant's compliance PDF.
- Cleared reliability-impacting technical debt across adjacent teams and codebases, increasing engineering output and velocity.

Projects

Latent-Structure Modeling of Student Outcomes (OULAD) *Research Project* [Report] 2026

- Modelled latent engagement and performance structure across 23,301 complete student-course registrations (of 25,820) via a parallel-analysis-validated 3-factor varimax model and 3-component PCA (72.9% variance); showed FA, GMM, and LDA converge on the same two latent dimensions.
- Benchmarked classifiers under 5-fold CV (logistic regression 67.0% vs. LDA 64.5% vs. 53.0% majority baseline) and showed LR lifts minority-class recall from 3.2% to 39.3%, diagnosing LDA's shared-covariance violation.

GPT from Scratch *Personal Project (in progress)* [Repo] 2026

- Building a character-level GPT in pure NumPy, growing one codebase from a bigram baseline into a full Transformer: tokenization, multi-head self-attention, Adam, and sampling.

Neural Network from Scratch (MNIST) *Personal Project* 2026

- Implemented a two-layer feedforward network for MNIST digit classification in pure NumPy, deriving and coding all gradient and backpropagation updates by hand.

Autonomous Planning & School Knowledge Agents *Personal Project* 2026

- Built an autonomous daily-planning agent (Claude Code, Obsidian) that regenerates a time-blocked calendar from a parsed task file and priority scheme, cutting daily planning from about 30 minutes to under 2.

- Built a companion school-workflow agent that auto-formats lecture and problem-set notes into a cross-linked knowledge base of theorem references, cutting note-cleanup and lookup time an estimated 40%.

F1 Race Winner Predictor *Personal Project* [Repo] 2024

- Built a supervised pipeline predicting Formula 1 race winners from driver and constructor rolling form, qualifying results, driver-track history, pit-stop reliability, and weather.
- Reached 40% race-winner and 60% podium accuracy on held-out races, well above the roughly 5% random and 30% pole-position baselines.

Plantinarium *6-person team, Java* [Repo] 2024

- Collaborated on an AI plant-identification desktop app (Java, Maven): photo upload to species identification, public/private galleries, likes, and notes; owned the account-management flow including secure account deletion with full data wipe.

Technical Skills

Languages: Python, R, SQL, Java, JavaScript, Bash, HTML, CSS
ML & Data: scikit-learn, NumPy, Pandas, SciPy, statsmodels, Matplotlib, Seaborn; neural nets and backprop implemented from scratch
Systems & Web: Airflow, Databricks, Postgres, GraphQL, React, Node.js, Flask, REST APIs, Maven, Linux/Unix
Tools: Git, GitHub, Jupyter, Excel, LaTeX
AI Tools: Claude, Claude Code, Cursor, LLM agents

Additional

StarCraft II Grandmaster (rank 100 NA, 5100 MMR); competitive Magic: The Gathering; Shotokan karate black belt.